Nikhitha Reddy Sama

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Education

Sacred Heart University — Fairfield, CT
Master of Science in Computer Science CGPA: 3.86/4 — *Aug 2023- Dec 2024*

Technical Proficiencies

Languages: Python, Java, C, C++, HTML, CSS, JavaScript, TypeScript, SQL; **Frameworks:** React, Vue, Node.js, Flask, Spring Boot **Databases:** MySQL, MongoDB, PostgreSQL, Redis, Oracle DB; **Cloud:** AWS (EC2, S3, Lambda, DynamoDB), Azure; **AI/ML**: NLTK, PyTorch; **DevOps Tools & Orchestration:** Git, GitHub Actions, Postman, Jira, Bitbucket, Jenkins, Docker, Kubernetes, Terraform, Apache (Kafka, Airflow), CI/CD; **Certifications:** Python Essentials, Programming Essentials in C, Programming Foundations with JavaScript, HTML, CSS; DAG Authoring (Airflow 3)

Work Experience

ABE Scott Enterprises — Herndon, VA
Software Engineer — *Aug 2025 - Present*

- Designed and deployed Java microservices in AWS and Azure, improving platform reliability by 35% and enabling seamless scaling.
- Enhanced trading dashboard accessibility with HTML, JavaScript, achieving 90% WCAG compliance and boosting user satisfaction.
- Automated CI/CD pipelines with Jenkins and Bash, cutting deployment times by 30% and improving release reliability.
- Implemented Docker and Kubernetes orchestration with proactive monitoring, reducing downtime by 25% during peak trading activity.

Sacred Heart University — Fairfield, CT
Graduate Research Assistant — *Apr 2024 – Jun 2024*

- Collected and analyzed bike safety data from 1M+ Reddit posts across 100+ subreddits using Reddit and Pushshift APIs, along with web scraping tools. Performed statistical analysis with Pandas and NumPy to extract actionable insights.
- Created interactive visualizations by leveraging Matplotlib and Seaborn to identify accident trends.
- Applied topic modeling and sentiment analysis to extract user concerns from large-scale text data. Fine-tuned Transformer based Large Language Models (LLM) on Google Colab using TensorFlow and CUDA GPUs, boosting training speed by 40%.

Accenture — Hyderabad, India
Software Engineer — *Oct 2022 – Aug 2023*

- Improved accessibility of the CAF online banking platform by leveraging HTML, CSS, JavaScript, and ARIA labels. Ensured WCAG compliance and ran accessibility and performance checks using Axe, WAVE, and Lighthouse.
- Streamlined Jenkins build workflows by automating micro-app dependency management with Bash scripts. This reduced build times and improved Continuous Integration and Continuous Deployment (CI/CD) reliability across environments.
- Improved backend reliability by debugging API failures and writing unit and integration tests in Postman.
- Refactored controller logic with asynchronous callbacks and multithreading, cutting API response times by 15%.
- Deployed Prometheus and Grafana to track payment API latency, enabling early detection of performance issues.

Associate Software Engineer — *Oct 2021 – Oct 2022*

- Implemented homepage functionalities in Java and REST APIs, applying stored procedures for backend data handling.
- Built responsive retail banking dashboards in React and JavaScript, delivering a seamless user experience across web and mobile. Resolved UI bugs and addressed cross-browser inconsistencies with Chrome Dev Tools.
- Migrated Kony online banking platform from monolithic to micro-app architecture, increasing system flexibility.
- Built scalable microservices in Spring Boot and implemented Redis caching to efficiently process millions of banking transactions with low latency and high reliability.
- Developed mobile-native UI for Euro Bank's account details page in React Native, optimizing performance with lazy loading and pagination, which reduced transaction retrieval time from 4 to 2 seconds and enhanced user satisfaction.
- Mentored interns on MVC architecture and supported Agile delivery by resolving 100+ JIRA tickets and 15+ user stories.

Projects

University Course Management System
Engineered a Python-based course management system using Red-Black Trees and Binary Min-Heaps, achieving O(log n) performance for course operations and supporting thousands of concurrent student registrations.

Nutritional Cost Analyzer
Built a full-stack web app (Flask, React, Oracle DB) to analyze food pricing vs. nutrition, improving analysis efficiency by 60% and enabling users to identify cost-effective, healthier options.